

Dylan Perdigão[✉]

dgp@dei.uc.pt

Francisco Antunes[✉]

fnibau@dei.uc.pt

Catarina Silva[✉]

catarina@dei.uc.pt

Bernardete Ribeiro[✉]

bribeiro@dei.uc.pt

CISUC/LASI

Department of Informatics Engineering

University of Coimbra

Coimbra – Portugal

Abstract

This paper proposes a population-coded convolutional spiking neural network for detecting financial fraud. By using the joint neural activity and optimizing hyperparameters through Bayesian optimization, our model achieves a recall of up to 47% at a 5% false positive rate, outperforming non-population coding SNNs. The approach delivers more consistent results and over 90% fairness in predictive equality across each sensitive attribute, such as age, income, and employment status. These results highlight the suitability of SNNs as a fair, performant, and energy-efficient alternative to traditional models. Future work will explore the integration of causal inference to enhance transparency for a more responsible AI.

1 Introduction

The rapid expansion of digital banking has increased both convenience for users and opportunities for fraudsters, making financial fraud detection a critical challenge. Traditional machine learning methods have been widely adopted for this purpose, but are often computationally expensive and energy-inefficient due to their reliance on large-scale floating-point operations. To address these limitations, spiking neural networks (SNNs) have emerged as a promising alternative [7], inspired by the brain's spike-based communication to deliver energy-efficient computation, particularly when deployed on neuromorphic hardware [2]. Beyond efficiency, SNNs can also benefit from population coding, a biologically inspired mechanism where groups of neurons collectively represent information. This approach enhances robustness and accuracy compared to the simplistic one-neuron-per-class design used in many classification tasks [3, 5]. In this research, we enhance traditional SNNs with population coding into a financial fraud detection task using the bank account fraud (BAF) suite of datasets [6]. The particularity of this benchmark is that our model is constrained to maintain a false positive rate below 5%, minimizing the impact on legitimate customers, while also ensuring fairness by avoiding biases related to age, income, or employment status. The results demonstrate the potential of population-coded SNNs to provide an energy-efficient, fair, and performant alternative to traditional fraud detection models.

2 Background

In spiking neural networks, neurons generate electrical impulses when the membrane potential U exceeds a threshold θ , driven by the intensity I over time t . The system is commonly modeled using the leaky integrate-and-fire (LIF) neuron [4], where the model treats the neuron as an electrical circuit composed of a capacitor C and a resistor R , defined by

$$\tau_m \frac{dU}{dt} = -U(t) + RI(t), \quad (1)$$

where $\tau_m = RC$ is the time constant. This differential equation can be approximated with the forward Euler method for computational use in sequential networks described as

$$U[t] = \underbrace{\beta U[t-1]}_{\text{decay}} + \underbrace{WX[t]}_{\text{input}} - \underbrace{S_{\text{out}}[t-1]\theta}_{\text{reset}}. \quad (2)$$

The neuron's potential evolves based on the decay β of past activity, weighted input signals WX , and a reset mechanism triggered by spiking events S_{out} , where the output is typically determined by counting spikes in the final layer [3]. Population coding extends this idea by assigning

multiple output neurons to each class, better reflecting biological neural activity and enhancing robustness in classification tasks.

3 Related Works

Fraudulent activities have become a significant concern for financial institutions. Several works have been proposed to improve fraud detection in this sector, namely with the BAF suite of datasets developed by Jesus et al. [6]. These datasets have the particularity of being composed of 1,000,000 instances with 30-32 features from which three are sensitive attributes (age, income, and employment status). Each dataset variant differs in its group size disparity for each attribute and is highly imbalanced towards fraud cases (approximately 1% of fraud). The Bayesian optimization of SNN parameters, explicitly using the LIF neuron model with convolutional architectures [8, 9, 11], or also tested other variants of the LIF neuron [10].

4 Proposed Approach

Our proposed approach¹ is built on a 1D-convolutional spiking neural network enhanced with population coding to detect bank account fraud. The network, illustrated in Figure 1, consists of convolutional layers followed by a fully connected layer mapped to a neuron population P , where half of the neurons are dedicated to fraudulent behavior and the other half to non-fraudulent ones. This setup enables the study of joint neural activity rather than single-neuron classification. The optimization process follows two stages. First, we apply multi-objective Bayesian optimization with a tree-structured Parzen estimator (TPE) sampler using Optuna [1], targeting maximization of true positive rate (TPR) and minimization of the false positive rate (FPR). Six configurations are explored by varying time steps of the spiking neuron ($S \in \{20, 50\}$) and the output population size ($P \in \{2, 20, 200\}$). Over 1,000 trials, key hyperparameters are tuned, including membrane decay and threshold values, surrogate gradient slope, and Adam optimizer parameters. Secondly, the best hyperparameters from each configuration are validated by running the model 100 times on the six variants of the BAF dataset [6], ensuring the robustness of the models.

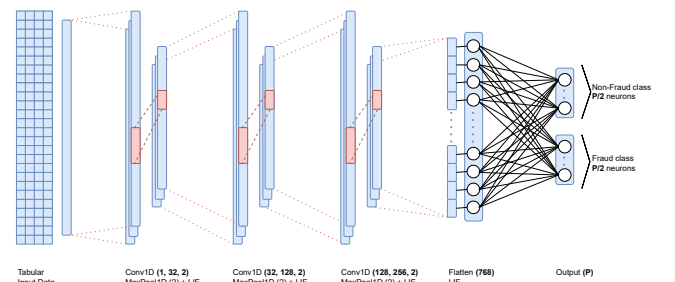


Figure 1: Convolutional spiking neural network with population coding mechanism.

¹Experimental work available on Github at <https://github.com/DylanPerdigao/Neural-Joint-Activity-in-SNNs>, and archived on Zenodo at <https://doi.org/10.5281/zenodo.13546088>

5 Results and Discussion

The Bayesian optimization shows that larger populations and higher time steps consistently improve TPR at the 5% FPR constraint. Full results are presented in Table 1. Configurations with $P = 2$ neurons perform worse, achieving between 24% to 34% of TPR, whereas populations of 20 reach up to 43% TPR with 50 time steps. The best result, 47.08% TPR, is achieved with $S = 50$ and $P = 200$, and is comparable to the LightGBM model in Jesus et al. [6]. Increasing the population of neurons reduces the variance, demonstrating higher and stable performance.

Fairness analysis further highlights the advantages of the proposed convolutional SNN compared to other machine learning (ML) models. Predictive equality across sensitive attributes (age, income, employment status) remains above 90%, in contrast to LightGBM, which exhibits more biases towards these sensitive attributes [6]. Additionally, higher population sizes show slight increases in the predictive equality.

Table 1: Performance and fairness results for SNNs with population P and time steps S .

Dataset	Model	TPR	FPR	PE _{age}	PE _{income}	PE _{employment}
Base	P2-S20	30.30 ± 11.45	2.48 ± 1.41	96.58 ± 2.67	96.81 ± 2.97	96.78 ± 2.89
	P2-S50	33.08 ± 13.14	2.84 ± 1.60	95.63 ± 4.28	96.71 ± 3.06	94.30 ± 5.41
	P20-S20	40.68 ± 5.78	3.60 ± 0.94	97.80 ± 1.94	98.05 ± 1.57	96.47 ± 2.64
	P20-S50	41.57 ± 6.15	3.90 ± 0.98	97.27 ± 1.40	98.15 ± 1.11	97.55 ± 1.89
	P200-S20	29.65 ± 11.11	2.67 ± 1.34	96.17 ± 3.60	96.59 ± 6.59	94.71 ± 5.99
	P200-S50	42.79 ± 0.59	4.07 ± 0.15	96.54 ± 2.98	98.02 ± 1.07	96.25 ± 3.47
Variant I	P2-S20	31.95 ± 11.67	2.93 ± 1.44	96.21 ± 3.43	97.09 ± 3.84	95.84 ± 3.31
	P2-S50	30.33 ± 11.19	2.59 ± 1.39	95.77 ± 4.52	97.13 ± 2.53	94.66 ± 5.92
	P20-S20	38.41 ± 5.68	3.56 ± 0.91	96.75 ± 2.55	98.06 ± 1.58	96.94 ± 2.00
	P20-S50	40.71 ± 3.94	4.01 ± 0.64	96.87 ± 2.43	98.33 ± 1.04	97.27 ± 2.13
	P200-S20	24.04 ± 11.45	2.29 ± 1.45	95.52 ± 4.46	96.11 ± 3.96	94.09 ± 5.73
	P200-S50	40.21 ± 4.87	3.95 ± 0.98	96.61 ± 2.38	98.34 ± 0.99	97.35 ± 1.41
Variant II	P2-S20	34.78 ± 8.98	2.94 ± 1.27	97.89 ± 1.56	97.27 ± 2.01	96.13 ± 3.45
	P2-S50	28.91 ± 14.28	2.20 ± 1.53	95.65 ± 5.38	96.15 ± 3.60	92.90 ± 9.26
	P20-S20	42.86 ± 5.63	3.77 ± 0.90	98.16 ± 1.33	97.73 ± 1.71	97.11 ± 2.10
	P20-S50	43.15 ± 5.02	4.03 ± 0.67	97.97 ± 1.11	98.25 ± 1.18	97.38 ± 1.70
	P200-S20	30.16 ± 10.61	2.78 ± 1.39	97.76 ± 2.11	97.42 ± 2.31	94.71 ± 4.64
	P200-S50	47.08 ± 2.85	4.32 ± 0.53	98.97 ± 0.34	98.67 ± 1.39	99.48 ± 0.19
Variant III	P2-S20	31.73 ± 8.33	3.24 ± 1.19	98.05 ± 1.45	97.77 ± 1.90	95.66 ± 3.30
	P2-S50	27.97 ± 9.77	2.57 ± 1.47	97.90 ± 1.51	97.15 ± 2.39	95.59 ± 3.81
	P20-S20	36.06 ± 4.93	3.60 ± 0.84	98.45 ± 1.14	98.12 ± 1.23	96.83 ± 2.12
	P20-S50	36.98 ± 6.19	3.90 ± 1.05	98.25 ± 1.48	97.36 ± 1.43	97.50 ± 1.86
	P200-S20	25.03 ± 8.92	2.62 ± 1.29	97.85 ± 1.75	97.09 ± 2.71	95.48 ± 3.95
	P200-S50	41.83 ± 0.78	4.85 ± 0.11	97.76 ± 1.16	98.93 ± 0.87	96.98 ± 3.21
Variant IV	P2-S20	25.26 ± 11.02	2.73 ± 1.61	97.83 ± 2.47	97.95 ± 2.52	96.62 ± 2.66
	P2-S50	24.40 ± 9.64	2.32 ± 1.31	97.38 ± 2.27	98.07 ± 1.61	94.79 ± 4.52
	P20-S20	32.99 ± 5.11	3.58 ± 0.92	97.77 ± 1.84	98.19 ± 1.51	97.06 ± 1.82
	P20-S50	34.79 ± 4.99	4.06 ± 0.89	98.54 ± 1.24	98.24 ± 1.73	96.66 ± 3.13
	P200-S20	25.43 ± 7.69	2.95 ± 1.27	97.56 ± 1.51	97.87 ± 1.67	96.60 ± 2.72
	P200-S50	35.54 ± 5.71	4.20 ± 1.05	98.68 ± 1.02	97.23 ± 1.32	95.58 ± 3.11
Variant V	P2-S20	34.96 ± 10.08	3.46 ± 1.32	97.70 ± 2.12	97.88 ± 2.20	95.40 ± 8.55
	P2-S50	28.84 ± 11.97	2.55 ± 1.39	93.68 ± 16.36	93.18 ± 17.49	92.95 ± 14.75
	P20-S20	36.41 ± 6.70	3.40 ± 1.00	98.03 ± 1.53	97.94 ± 1.49	96.47 ± 2.48
	P20-S50	39.48 ± 7.27	3.96 ± 1.07	98.47 ± 1.16	98.38 ± 1.61	96.17 ± 2.05
	P200-S20	24.97 ± 11.45	2.55 ± 1.50	97.37 ± 2.50	96.87 ± 3.90	95.72 ± 4.38
	P200-S50	43.10 ± 2.83	4.58 ± 0.53	99.31 ± 0.56	98.94 ± 0.65	97.80 ± 1.59

6 Conclusion and Future Work

This work studied 1D-convolution SNNs with population coding for fraud detection, demonstrating that the best configuration achieved a 47% TPR at a 5% FPR, outperforming non-population coding SNNs while ensuring over 90% fairness across sensitive attributes. The approach not only improves detection robustness but also avoids bias, making it suitable for real-world deployment. Additionally, SNNs hold promise for sustainable AI due to their energy efficiency compared to traditional artificial neural networks.

Future research will explore the integration of causal inference into SNNs to capture the relationships between spiking neurons, thereby enhancing model interpretability. This will align with regulatory demands and increase public trust by moving towards more transparent and explainable AI.

Acknowledgements

This work is funded by national funds through the FCT – Foundation for Science and Technology, I.P., within the scope of the research unit UID/00326 – Centre for Informatics and Systems of the University of Coimbra, and by the Portuguese Recovery and Resilience Plan (PRR) through project C645008882-00000055, Center for Responsible AI. FCT also supports this research through the Research Grant 2024.01388.BD.

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